Uniformly Accelerated Motion

Kinematics of straight-line motion with constant acceleration · problem set 1 · Dad's copy: tags, answers, references

The rules of the game

For **every** problem, including the warm-up:

1. Choose the positive direction and set $x_0 = 0$ at the start.
2. Solve for what is asked.
3. Draw the three graphs against time: $a(t)$, $v(t)$, $x(t)$ — axes labelled, key values marked (initial/final speeds, times, distances).

Graphs need not be to exact scale, but the *shapes* must be right: with constant acceleration $a(t)$ is a horizontal line, $v(t)$ is a straight line with slope a, and $x(t)$ is a parabola (bending up if $a > 0$, down if $a < 0$). Use $g = 10$ m/s^2.

The toolbox — four equations, each *missing* one quantity:

	Equation	Missing	Use when. . .
(i)	$v = u + at$	s	distance does not matter
(ii)	$s = ut + \frac{1}{2}at^2$	v	final speed does not matter
(iii)	$v^2 = u^2 + 2as$	t	time does not matter
(iv)	$s = \frac{(u+v)}{2}t$	a	acceleration does not matter

u — initial velocity, v — final velocity, a — acceleration (constant), s — displacement, t — time. Each problem hands a different combination of these and asks for the rest. **Only in this copy:** the "Missing" column and the [given → find] tags on the problems — working that out himself is part of the training.

Problems

0. Warm-up (done in class — now add the graphs). A train enters a station at speed v. The station has length L, and the train stops exactly at its far end, decelerating uniformly. Treating the train as a point, find the acceleration and the stopping time. [given u, v=0, s → a, t]

1. Away from the lights. A car starts from rest and accelerates uniformly at 2 m/s^2 for 6 s. Find its final speed and the distance covered. [given u=0, a, t → v, s]

2. Between stations. A train speeds up from 10 m/s to 25 m/s in 30 s at constant acceleration. Find the acceleration and the distance travelled. [given u, v, t → a, s]

3. The cyclist. A cyclist moving at 6 m/s accelerates uniformly at 2 m/s^2 along a 16 m stretch of road. Find his speed at the end of the stretch and the time it took. [given u, a, s → v, t]

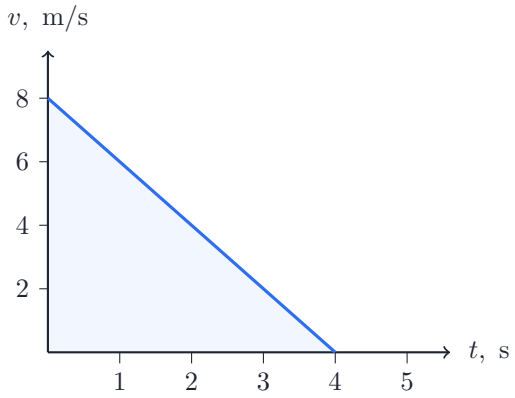
4. Braking distance. A car travels at 20 m/s. The driver brakes, and the car decelerates uniformly at 5 m/s^2. (a) How long and how far does it travel before stopping? (b) If the initial speed were doubled (40 m/s), how far would it travel? How many times longer is that? [given u, v=0, a → t, s]

5. **Off the bridge.** A stone is released from rest from a bridge and hits the water after 2 s. How high is the bridge, and how fast does the stone hit the water? (Ignore air resistance.) [given $u{=}0$, $a{=}g$, $t \rightarrow v$, s]

6. **Straight up.** A ball is thrown straight up at 15 m/s. (a) How long does it take to reach the top? (b) What is the maximum height? (c) When does it return to the hand, and with what velocity? — On your graphs, pay special attention: what is the acceleration *at the very top*? [given u, $a{=}-g \rightarrow t$, s, symmetry]

7. **The whole trip (graphs first!).** A bus leaves a stop: it accelerates uniformly from rest to 12 m/s in 4 s, drives at a steady 12 m/s for 6 s, then brakes uniformly to a stop in 3 s. Draw all three graphs *first*; then find the acceleration and the distance for each phase, the total distance, and the average speed of the trip. [three phases → piecewise graphs]

8. **Read the graph.** The v–t graph of a tram is shown. From the graph, find the initial speed, the acceleration (slope), and the distance travelled (area under the graph). Then sketch the matching $a(t)$ and $x(t)$. [given graph → u, a, s]



9. **Challenge: where equation (iv) comes from.** Using a v–t graph for uniform acceleration, show that the displacement equals the area of a trapezium: $s = \frac{1}{2}(u+v)\,t$. Then use this to re-derive your answer to Problem 0 in one line. [derivation]

Answers

0. $|a| = \dfrac{v^2}{2L}$ directed against the motion; $t = \dfrac{2L}{v}$.

1. $v = 12$ m/s; $s = 36$ m.

2. $a = 0.5$ m/s^2; $s = 525$ m (via eq. (iv): 17.5×30).

3. $v = \sqrt{6^2 + 2 \cdot 2 \cdot 16} = 10$ m/s; $t = 2$ s.

4. (a) $t = 4$ s, $s = 40$ m; (b) 160 m — 4× longer: braking distance grows as v^2 (eq. (iii)).

5. $h = \frac{1}{2} \cdot 10 \cdot 2^2 = 20$ m; $v = 20$ m/s.

6. (a) 1.5 s; (b) 11.25 m; (c) returns at 3 s with 15 m/s downward. At the top $v = 0$ but $a = -g = -10$ m/s^2 — the $a(t)$ graph is one unbroken horizontal line.

7. $a_1 = +3$ m/s^2, $s_1 = 24$ m; $a_2 = 0$, $s_2 = 72$ m; $a_3 = -4$ m/s^2, $s_3 = 18$ m. Total 114 m in 13 s; average speed ≈ 8.8 m/s. The $v(t)$ graph is a trapezium; its area is the total distance.

8. $u = 8$ m/s; $a = -2$ m/s^2; $s = \frac{1}{2} \cdot 8 \cdot 4 = 16$ m (the shaded triangle). $x(t)$ is a parabola bending down, flattening out at $t = 4$ s.

9. Trapezium with parallel sides u and v, width t: area $= \frac{1}{2}(u+v)t$. For the train: $L = \frac{1}{2}(v+0)\,t \Rightarrow t = 2L/v$, then $a = (0 - v)/t = -v^2/(2L)$. Same answers, no quadratic equations.

Graph checklist for marking: $a(t)$ horizontal (negative where braking); $v(t)$ straight, crossing zero exactly at the stopping/turn-around moment; $x(t)$ a parabola with the right bend, horizontal tangent where $v = 0$, never jumping.

More practice from our archive

- **Siyavula, Physical Sciences Gr 10** — `docs/textbook/siyavula-physical-sciences/Gr10_PhysicalSciences_Learner_Eng.pdf`
  ch 21 *Motion in one dimension*: worked examples throughout (PDF pp. 403–452, graph-heavy from p. 430) and *End of chapter exercises* (PDF pp. 453–457) — includes drawing and reading x–t/v–t graphs. Brief solutions in ch 25 ($\approx$ PDF p. 504).

- **NCERT, Class 9 Science** — `docs/textbook/ncert-class9-physics/ch04-describing-motion.pdf`
  §4.2 graphs of motion (PDF pp. 9–16), §4.3 kinematic equations (PDF pp. 16–18), exercises *Revise, Reflect, Refine* (PDF pp. 21–24, graph question on p. 23).

- **CK-12, Quizzes & Tests** — `docs/textbook/ck12-physical-science/CK-12-Physical-Science-MS-Quizzes-a`
  `pdf`
  quiz 12.2 *Speed and Velocity* (PDF p. 161), quiz 12.3 *Acceleration* (PDF p. 164), Unit-4 *Motion and Forces* test (PDF p. 353); answers are in the same file.